Bench exercise: mark out a gooseneck before you cut

Japanese Joinery by Hand · Episode 10. Mark a full koshikake kama-tsugi 腰掛け鎌継ぎ on two pieces of softwood, in the training course's order, and check every line before any cutting. Cutting is optional: do it with a teacher if you can. About 60 minutes for the marking.

> **Safety.** If you go on to cut: clamp the work, keep both hands behind the edge, and stop saw cuts at the line. Sharp-tool technique, and how to use these tools safely, is best learned from a teacher, at a bench, in person.

1 · Prepare

1. Two planed softwood blocks of the same square section, each about 400 mm long. The episode draws 120 mm; 75–90 mm is fine. Scale the joint to suit: at 90 mm, a neck about 22 mm wide and a head and neck about 110 mm long.
2. Write "male" (upper piece) and "female" on them. Strike center lines on the top faces, as in Episode 1.

2 · Mark the female (the course's order)

1. The end line, then the seat line.
2. Lines across for the neck and the head's tip, then the cut line.
3. The neck lines, split evenly from the center line.
4. The head's slanted lines, from the tip toward the neck.
5. The seat lines down both side faces.

3 · Mark the male the same way

1. End line (the head's tip), the neck and tip lines and the cut line, the neck lines from the center, the slanted head lines, and the seat lines down both sides.

4 · Check before cutting

- Is the male's head exactly the same as the socket drawn on the female, measured from each center line?
- Can you say, for each cut, how much of the line you will leave? The course leaves half the line on most cuts and takes it at the socket's tip.
- Where will the entry taper go (line left at the top, taken at the bottom on the head's and neck's sides)?

What you should have

- Two fully marked pieces you could hand to someone else to cut.

The sequence follows one training course (ものつくり大学 WBT); its exercise adds a stub tenon at the seat, left out here. The scaled sizes keep Episode 9's test-specimen proportions (neck ≈ W/4, head and neck ≈ 1.25 W, flare ≈ L/20 a side); they are not a carpentry standard.